

MIAE 215 · Mini-Course Lesson 4

Expressions, Operators, Precedence & Modulo Tricks

Concordia University · Department of Mechanical, Industrial & Aerospace Engineering (MIAE)

Source: [Lesson 4 Web Page](#) | Video: [lesson4_expressions.mp4](#)

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1. Operators, Operands & Expressions

- **Operator:** Performs an action on data (e.g., `+`, `-`, `*`, `/`, `%`).
- **Operand:** The data items or variables acted upon by an operator.
- **Expression:** A combination of operators and operands evaluating to a single value.

```
y = x + 1 + z * w; // Expression containing 4 operators and multiple operands
```

2. Sequential Assignment (=) Lifecycle

In C++, assignment `=` is imperative:

```
x = 3;  
x = 2 * x + 1; // Evaluates RHS: 2*(3)+1 = 7, then stores 7 in x
```

This is **sequential memory mutation**, not a simultaneous algebraic equation.

Chained Assignments

Assignment associates right-to-left:

```
z1 = z2 = z3 = 11.0; // z3 gets 11.0, then z2 gets 11.0, then z1 gets 11.0
```

3. The Modulus Operator (%) & Digit Peeling

The `%` operator returns the integer remainder after division: $237 \% 10 = 7$

Extracting Digits Right-to-Left:

```
int num = 237;
int d1 = num % 10; // d1 = 7
num = num / 10;    // num = 23
int d2 = num % 10; // d2 = 3
num = num / 10;    // num = 2
int d3 = num % 10; // d3 = 2
```

4. Operator Precedence & Associativity Hierarchy

Precedence Hierarchy:

1. () Parentheses
2. ++, -- (prefix / postfix)
3. *, /, % (multiplicative)
4. +, - (additive)
5. = (assignment)

To evaluate rational expressions like $y = \frac{ax^2+bx+c}{dx+e}$, **parentheses are mandatory**:

```
y = (a*x*x + b*x + c) / (d*x + e);
```

5. Teacher Code & Exercise Walkthroughs

- [lesson4.cpp](#) : Comprehensive lesson code covering arithmetic operators and precedence.
- [lesson4_exercises.cpp](#) & [lesson4_exercises_solutions.cpp](#) : Exercise code and solutions.
- [lesson4_more.cpp](#) : Supplemental practice exercises.